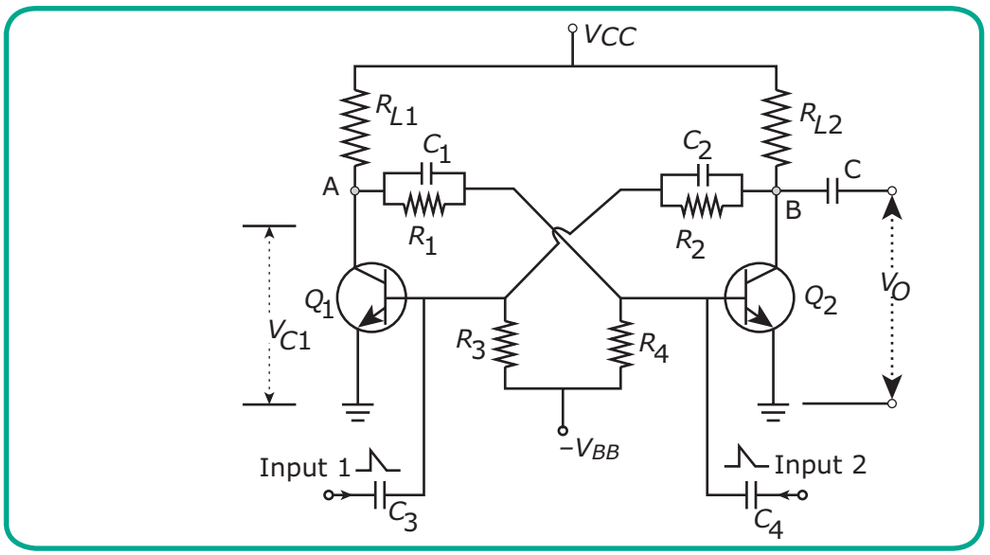


Figure 7.9 **Bistable Multivibrator**

transistor Q_1 through a resistor R_2. Similarly, the output of Q_1 is coupled to the base of Q_2 through resistor R_1. The main purpose of capacitors C_1 and C_2 is to improve the switching characteristic of the circuit by passing the high frequency components of the square wave pulses. This allows fast rise and fall times, so that these square waves will not be distorted. C_1 and C_2 are called commutating capacitors or speed up capacitors.

Working Principle: When the circuit is switched-ON, one of the transistors will start conduct slightly higher than the other. This transistor is thus driven into saturation (i.e. ON). Then, because of the regenerative feedback action, the other transistor is taken into cut-off (i.e. OFF) state. Let us consider transistor Q_1 is ON and Q_2 is OFF. It is a stable state of the circuit and will remain in this state till a trigger pulse is applied from outside. A positive triggering pulse applied to the reset input (base of Q_2) increases its forward bias, thereby turning transistor Q_2 ON, and there is an increase in collector current and decrease in collector voltage occur. The fall in the collector voltage is coupled to the base of Q_1, which in turn turned OFF. The circuit is then in its second stable state until a positive trigger pulse is applied to the base of Q_1.

A similar action can be achieved by applying a negative pulse at the set input for transition from the first stable state to the second stable state and by applying a negative pulse at the reset input, reverse transition can be obtained.

Applications

1. It is used in computer memory circuits.
2. It is used as memory element in shift registers, counters and so on.
3. It can also be used as a frequency divider.



Uses of multivibrator

- They are used as a frequency divider
- Used as sawtooth generators
- They are used as wave and pulse generator.
- They are used as standard frequency source.
- They are used in Radar and TV circuits
- They are also used as memory elements in computer.

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